

Recursive Nearby-Factorization and Quotient-Product Descent

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30-08-2026

Abstract

This paper develops a recursive viewpoint on the factorization of an integer $n = pq$ using auxiliary factorizations of nearby integers $n + x$. The central construction is the quotient-product relation induced by two moduli r_1, r_2 . Writing

$$p = a + kr_1, \quad q = b + \ell r_2,$$

leads to the exact quantity

$$T = \left\lfloor \frac{n}{r_1 r_2} \right\rfloor, \quad E = T - k\ell,$$

and hence

$$K = k\ell = T - E.$$

The important observation is that K is a much smaller integer than n in the parameter regime of interest. If an auxiliary integer $n + x$ is easier to factor, then its factorization can reveal its own quotient coordinates (k_x, ℓ_x) and therefore its smaller quotient product $K_x = k_x \ell_x$. This creates a recursive chain in which a large factorization problem is related to a sequence of smaller arithmetic problems.

A further simplification is obtained by choosing x so that $n + x$ is even. For odd semiprime n , an odd x gives

$$n + x = 2m,$$

so at least one prime factor is known immediately. More generally, choosing x such that $n + x$ has a prescribed collection of small factors can produce an auxiliary number whose remaining cofactor is substantially smaller than the original integer.

The purpose of this work is not to claim a complete factoring algorithm, but to formalize the recursive mechanism, identify the information carried by E_x , K_x , k_x , and ℓ_x , and state the precise reconstruction problem that a successful recursive method would have to solve.

1 Introduction

The starting problem is the factorization of a composite integer

$$n = pq,$$

where p and q are unknown factors. Earlier investigations considered representations based on two moduli and studied whether modular information could directly identify p and q . These investigations showed that congruences such as

$$pq \equiv n \pmod{r_1 r_2}$$

are primarily relational: once one factor is supplied, the other is immediately determined modulo the chosen modulus. The difficult part is therefore not verification, but generation of a useful candidate location.

The present approach changes the viewpoint. Instead of attempting to extract p and q directly from one representation of n , we introduce nearby integers $n + x$ and attempt to factor those auxiliary integers first. The resulting factors define smaller quotient coordinates and, in turn, smaller quotient products. This creates a possible recursive descent.

The principal chain is

$$n \longrightarrow n + x \longrightarrow (p_x, q_x) \longrightarrow (k_x, \ell_x) \longrightarrow K_x = k_x \ell_x.$$

The research question is whether enough information survives this chain to reconstruct the factor coordinates associated with the original n .

2 Two-Modulus Quotient Coordinates

Let r_1, r_2 be positive integers, typically prime and chosen so that $\gcd(r_1, r_2) = 1$. Write the unknown factors in quotient-remainder form:

$$p = a + kr_1, \tag{1}$$

$$q = b + \ell r_2, \tag{2}$$

with

$$0 \leq a < r_1, \quad 0 \leq b < r_2.$$

Substitution into $n = pq$ gives the exact expansion

$$n = k\ell r_1 r_2 + a\ell r_2 + bkr_1 + ab. \tag{3}$$

Define

$$T = \left\lfloor \frac{n}{r_1 r_2} \right\rfloor \tag{4}$$

and

$$E = T - k\ell. \tag{5}$$

Then the smaller quantity

$$K = k\ell = T - E \tag{6}$$

is an exact integer representation of the quotient coordinates.

The important point is the scale separation. When $r_1 r_2$ is large relative to the relevant factors, k and ℓ can be much smaller than p and q , and consequently $K = k\ell$ can be much smaller than n . This does not solve the original problem by itself, because E is unknown. It does, however, identify a potentially useful intermediate problem: recover K , factor K , and use its divisor pairs as candidates for (k, ℓ) .

3 The Defect E

Equation (3) gives

$$n - k\ell r_1 r_2 = a\ell r_2 + bkr_1 + ab.$$

After division by $r_1 r_2$, the integer part of the right-hand side creates the defect between T and $k\ell$. A useful exact decomposition is obtained by writing

$$a\ell = c_1 r_1 + d_1, \quad 0 \leq d_1 < r_1, \tag{7}$$

$$bk = c_2 r_2 + d_2, \quad 0 \leq d_2 < r_2. \tag{8}$$

The resulting carry identity has the form

$$E = c_1 + c_2 + c_3, \tag{9}$$

where c_3 is the final carry generated by the remaining residue terms. In particular,

$$c_1 = \left\lfloor \frac{a\ell}{r_1} \right\rfloor, \quad c_2 = \left\lfloor \frac{bk}{r_2} \right\rfloor.$$

This decomposition is valuable because it demonstrates that E is not an arbitrary error term. It is an exact bookkeeping quantity generated by the cross terms of the product expansion.

Nevertheless, knowledge that E has internal structure is not equivalent to knowing E from n alone. The recursive strategy described below is therefore designed to obtain new information from related integers rather than relying exclusively on a direct inversion of E .

4 Nearby Integers and the E_x Construction

For an integer displacement x , define

$$n_x = n + x.$$

Suppose n_x admits a factorization

$$n_x = p_x q_x.$$

Using the same two moduli, write

$$p_x = a_x + k_x r_1, \tag{10}$$

$$q_x = b_x + \ell_x r_2. \tag{11}$$

Define

$$T_x = \left\lfloor \frac{n_x}{r_1 r_2} \right\rfloor, \tag{12}$$

$$E_x = T_x - k_x \ell_x, \tag{13}$$

$$K_x = k_x \ell_x = T_x - E_x. \tag{14}$$

Thus every nearby factorization provides a complete quadruple

$$(T_x, E_x, k_x, \ell_x),$$

and the smaller product

$$K_x = k_x \ell_x.$$

This is fundamentally different from using $n + x$ merely as another integer to factor. The nearby factorization is converted into a lower-scale representation that is directly comparable with the original quotient coordinates.

5 Exact Relation Between E_x and K_x

There is an immediate identity relating the original and nearby states:

$$E_x - E = (T_x - T) - (K_x - K). \tag{15}$$

This equation is exact.

If the quotient T does not change between n and $n + x$, then

$$T_x = T,$$

and therefore

$$E_x - E = -(K_x - K). \tag{16}$$

Consequently,

$$E_x = E \iff K_x = K$$

within any region where $T_x = T$.

This observation is important for the recursive proposal. It shows that E_x does not need to be treated as an isolated mysterious quantity. Once the nearby factorization gives K_x , its relationship to T_x produces E_x automatically.

6 Why Choose $n + x$ to Be Even?

Assume the original integer n is odd, as occurs when p and q are odd primes. If x is odd, then

$$n + x$$

is even. Hence

$$n + x = 2m, \quad m = \frac{n + x}{2}. \quad (17)$$

The prime factor 2 is therefore known without any search.

More generally, if x is chosen so that

$$n + x = 2^t u,$$

then all factors of 2^t are known immediately and only the odd cofactor u remains to be factored.

The recursive viewpoint is not restricted to powers of two. One can seek auxiliary displacements for which

$$n + x = s u$$

with s having a known factorization composed of small primes. Then the remaining factorization problem concerns the smaller cofactor u .

The crucial distinction is that the auxiliary integer is not expected to be useful merely because it is numerically close to n . It is useful when its factorization is deliberately easier or contains known structure.

7 The Recursive Descent Hypothesis

The proposed recursive process can be expressed as follows.

7.1 Level 0: Original Integer

Start with

$$n = pq.$$

The factors p, q are unknown.

7.2 Level 1: Nearby Auxiliary Integer

Choose x so that $n + x$ is easier to factor. Compute

$$n + x = p_x q_x.$$

The auxiliary factors define

$$k_x = \left\lfloor \frac{p_x}{r_1} \right\rfloor, \quad \ell_x = \left\lfloor \frac{q_x}{r_2} \right\rfloor.$$

Hence

$$K_x = k_x \ell_x.$$

7.3 Level 2: Smaller Quotient Product

The number K_x is much smaller than the original n in the intended regime. Therefore we factor K_x directly:

$$K_x = d_1 d_2.$$

Every divisor pair produces a candidate quotient pair

$$(k_x, \ell_x) = (d_1, d_2).$$

The quotient coordinates then define intervals for the corresponding factor residues:

$$d_1 r_1 \leq p_x < (d_1 + 1) r_1, \tag{18}$$

$$d_2 r_2 \leq q_x < (d_2 + 1) r_2. \tag{19}$$

7.4 Level 3 and Beyond

The same process may be applied recursively to the smaller objects that arise from the quotient coordinates or from related nearby factorizations. Schematically,

$$n \rightarrow n + x_1 \rightarrow K_1 \rightarrow n + x_2 \rightarrow K_2 \rightarrow \cdots. \tag{20}$$

The hypothesis is that repeated descent eventually reaches integers small enough that their complete factorization is inexpensive. Their recovered quotient structure is then propagated upward.

8 What Information Must Propagate Upward?

The recursive proposal is only useful if information from a lower level can constrain a higher level. Several quantities are available for this purpose:

- (1) The exact lower-level quotient product K_x .
- (2) The divisor pairs of K_x .
- (3) The quotient coordinates (k_x, ℓ_x) .
- (4) The defect $E_x = T_x - K_x$.
- (5) The carry decomposition of E_x .
- (6) The quotient-cell locations implied by (k_x, ℓ_x) .
- (7) Relations between nearby states when T_x remains constant.

The essential reconstruction question is therefore not *whether* K_x can be factored. It is whether the resulting divisor pairs provide a sufficiently small set of possibilities for the original (k, ℓ) .

9 The Role of Stability

Nearby factorizations can sometimes remain in the same quotient cells. In that case

$$k_x = k, \quad \ell_x = \ell,$$

which immediately gives

$$K_x = K.$$

This is a geometric stability phenomenon.

A stronger and more interesting possibility is a cross-cell collision:

$$(k_x, \ell_x) \neq (k, \ell)$$

but

$$k_x \ell_x = k \ell.$$

Such a relation would show that the smaller product K can remain stable even while the individual quotient coordinates move.

These two cases must be distinguished carefully. Same-cell stability is expected from quotient geometry; cross-cell equality would provide additional arithmetic structure.

10 Candidate Reconstruction of the Original Factors

Suppose a recursive branch has produced a candidate original quotient pair (k, ℓ) . Then the unknown residues satisfy

$$p = kr_1 + a, \quad q = \ell r_2 + b,$$

with

$$0 \leq a < r_1, \quad 0 \leq b < r_2.$$

The exact product equation becomes

$$n = (kr_1 + a)(\ell r_2 + b). \tag{21}$$

The recursive information reduces the problem to finding a compatible residue pair (a, b) inside a bounded rectangle.

The carry identity provides additional constraints. In particular,

$$c_1 = \left\lfloor \frac{a\ell}{r_1} \right\rfloor, \quad c_2 = \left\lfloor \frac{bk}{r_2} \right\rfloor,$$

and

$$E = c_1 + c_2 + c_3.$$

Thus a recovered value of K narrows (k, ℓ) , while a recovered value of E narrows the admissible carry states.

The intended chain is therefore

$$K \rightarrow (k, \ell) \rightarrow (c_1, c_2, c_3) \rightarrow (a, b) \rightarrow (p, q).$$

11 Why the Recursive Viewpoint Is Different

The earlier modular experiments primarily used congruences as filters for already-generated candidates. In contrast, the recursive hypothesis tries to change the source of the candidates themselves.

The distinction can be summarized as

filtering approach:	p -candidates $\rightarrow$ modular tests,
recursive approach:	$n + x \rightarrow$ smaller factorization $\rightarrow K_x \rightarrow$ quotient candidates.

The latter can only be considered a genuine reduction if the auxiliary factorizations are collectively cheaper than factoring n directly and if the information they provide can be propagated upward without recreating the original search space.

12 The Central Bottleneck

There are two distinct unknowns in the recursive construction.

First, one must choose useful displacements x . An arbitrary nearby number is not necessarily easier to factor. The auxiliary search therefore needs a principled way of finding displacements for which $n + x$ has favorable factor structure.

Second, even when $n + x$ has been successfully factored, the resulting K_x must identify useful information about the original factorization. Knowing a small integer is not automatically equivalent to knowing which of its divisor pairs corresponds to the original quotient coordinates.

Thus the recursive method must eventually establish a statement of the form

$$\text{easy auxiliary factorization} \implies \text{small candidate set for } (k, \ell)$$

and then

$$\text{small candidate set for } (k, \ell) \implies \text{small candidate set for } (p, q).$$

13 A Precise Recursive Research Program

A complete experimental program based on this hypothesis should treat the following as separate stages.

13.1 Auxiliary Selection

Find x for which $n + x$ has a known or deliberately favorable factor structure, beginning with the guaranteed factor 2 obtained from odd x when n is odd.

13.2 Auxiliary Factorization

Factor the remaining cofactor of $n + x$ and record the exact tuple

$$(T_x, E_x, k_x, \ell_x, K_x).$$

13.3 Quotient-Product Factorization

Factor K_x and enumerate its divisor pairs. Since K_x is much smaller than n , this is the intended lower-level factoring step.

13.4 Upward Reconstruction

Use the candidate quotient coordinates, carry constraints, and residue relations to generate candidate factors for the original n .

13.5 Recursive Propagation

Apply the same mechanism to smaller auxiliary quantities whenever their factorization exposes another quotient-product representation.

14 Necessary Conditions for a Successful Method

A successful recursive mechanism would need to satisfy all of the following conditions simultaneously:

- (1) The chosen $n + x$ values must be significantly easier to factor than n itself.
- (2) Their factorizations must reveal exact lower-level quotient data.
- (3) Factoring the corresponding K_x values must produce a small collection of plausible quotient pairs.
- (4) The ambiguity among divisor pairs must shrink rather than remain proportional to the original search space.
- (5) The reconstruction of (p, q) must not require enumerating the original factor interval.
- (6) The recursive depth needed to reach easily factorable values must remain computationally manageable.

Failure of any one of these conditions can turn the apparent recursion into another representation of the same search problem.

15 Discussion

The new viewpoint does not rely on the assumption that a single modular congruence directly exposes the hidden factors. Instead, it treats factorization as a hierarchy of related problems.

The basic hierarchy is

$$n \longrightarrow n + x \longrightarrow (k_x, \ell_x) \longrightarrow K_x \longrightarrow \text{factor}(K_x).$$

The hope is that each step reduces the effective size of the unknown object and that the resulting information can be propagated back upward.

The special choice of even neighbors is particularly attractive because it provides an immediate nontrivial factor. For odd n , every odd displacement x gives an even auxiliary integer. This creates a systematic family of related factorizations rather than requiring an arbitrary second composite.

However, the decisive mathematical question remains whether the recursive information is genuinely new. In particular, if obtaining and processing the lower-level states implicitly requires a search whose cost scales like the original factorization problem, then the method has changed the representation without changing the underlying difficulty.

The purpose of the recursive framework is therefore to locate a possible information bottleneck that is different from direct candidate filtering: can deliberately easy factorizations of related integers reveal the hidden coordinates of the original factorization?

16 Conclusion

The two-modulus quotient representation introduces a small intermediate quantity

$$K = k\ell = T - E,$$

which is potentially far smaller than the original composite n . The defect E has an exact carry decomposition, but direct inversion of that decomposition does not automatically identify K .

The recursive nearby-factorization hypothesis proposes another route. By factoring suitably chosen integers $n + x$, one can obtain exact values T_x , E_x , k_x , and ℓ_x , and hence the smaller product

$$K_x = k_x\ell_x = T_x - E_x.$$

The factorization of K_x can then generate candidate quotient coordinates. If suitable auxiliary integers can be found recursively, the factorization of the original n may be decomposed into a sequence of smaller factorization problems.

Choosing $n + x$ to be even gives an especially simple first step:

$$n + x = 2m,$$

with the factor 2 known immediately. More generally, auxiliary values with intentionally favorable small-factor structure may create deeper recursive reductions.

At present, this framework should be regarded as a precise research hypothesis rather than a proof of a new factoring algorithm. The essential future test is whether the information obtained from easier auxiliary factorizations produces a genuinely shrinking candidate set for the original (p, q) without recreating the original search.